

Patient Shortage in Histopathology: Do Additional Patients Help Without Adding Coverage?

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Abstract

At a fixed patch budget, TCGA-UT patch classifiers trained on more patients are more accurate than redundancy-adjusted effective support predicts. Coverage of tissue source sites explains little of this residual, and a first test of patient coverage stopped before training because its low-coverage allocation covered held-out patients almost as well as a random one. This experiment repeats the test with a low-coverage allocation built around the coverage measure. Patient similarity is the cosine distance between patient-mean Virchow2 embeddings, so all 30 cancer types enter. Three nested allocations share a budget of 160 patches per class: five random anchor patients with 32 patches each; the anchors and the fifteen nearest neighbours of a single *focal* anchor with eight patches each; and the anchors and fifteen randomly drawn patients with eight patches each. Concentrating all additions around one anchor leaves the nearest training patient of most held-out patients unchanged, so the first step quadruples patients while adding little coverage, and the second step adds coverage while patients, patches, and effective support stay fixed. The *coverage gain* compares the two broad allocations, and the *within-concentration residual* measures what the first step adds beyond effective support. A prespecified manipulation check, run before any classifier is trained, requires the concentrated allocation to attain at most half of the random allocation's coverage improvement. A coverage gain above one percentage point with a residual below one point per doubling of patients would identify patient coverage as the missing source of information; the reverse pattern would locate it in properties of additional patients that their mean embedding does not express. The concentrated allocation attains 67 to 73 percent of the random allocation's coverage improvement in the three splits, and only testicular germ cell tumours meet the requirement, so the experiment stops at the manipulation check and both contrasts remain unmeasured. A diagnostic on the same embeddings shows why: for more than half of the held-out patients whose nearest anchor is not the focal anchor, a neighbour of the focal anchor lies closer than their own anchor, and these patients supply about twice as much of the leakage as the focal cell. Five further nested rules, including coverage-maximizing farthest-point and facility-location selection, reach leakage no lower than 0.53, because quadrupling patients shortens the nearest-patient distance for most held-out patients whichever patients are added. Coverage still varies strongly between cohorts of equal size, however: twenty patients forming one neighbourhood and twenty patients selected by facility location around the same anchor differ in coverage by 84 to 95 percent of the change from five to twenty random patients. A follow-up experiment therefore compares clustered, random, and facility-location allocations at a fixed patient count to distinguish concentration damage from improvement over ordinary sampling.

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1 Motivation

Spreading a labelling budget over more patients improves TCGA-UT patch classification beyond what redundancy explains. At 160 patches per class, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 9.52 percentage points of patient-macro balanced accuracy (Queisler, 2026a). *Effective support*, the patch count discounted for within-patient redundancy, accounts for less than half of this advantage: at equal effective support, each doubling of patients still adds 2.62 points (Queisler, 2026b), about 5.2 of the 9.52 points over the two doublings of this comparison. Tissue source sites explain little of this residual: doubling sites at a fixed patient count adds 1.06 points, and twenty patients from five sites are at least as accurate as twenty randomly drawn patients (Queisler, 2026d). Most of the residual lies in the seventeen smaller cancer types, which the site design could not reach.

Patient coverage, the share of a cancer type’s between-patient variation that the training patients represent, is the remaining candidate that requires no labels (Komura et al., 2022). Its first test compared a broad allocation formed by the nearest neighbours of five anchor patients with one formed by randomly drawn patients (Queisler, 2026c). The test stopped at its prespecified manipulation check, before any classifier was trained: the neighbour allocation reached 84 to 88 percent of the random allocation’s coverage improvement. The cause lies in the coverage measure, the mean distance from held-out patients to their *nearest* training patient. Because the anchors are part of both allocations and the neighbours were spread over all five of them, both rules shortened this distance for nearly the same held-out patients. A low-coverage allocation must therefore leave the nearest training patient of most held-out patients unchanged, not merely add patients that resemble the anchors.

Concentrating all additional patients around a single anchor meets this requirement. The four other anchors keep their neighbourhoods untouched, so coverage can improve only for held-out patients near the chosen anchor. The question stays the one posed by Queisler (2026c): *does a training patient unlike the current cohort contribute more than a patient resembling it, and do additional patients still help beyond effective support when they add little coverage?*

2 Study design

Only the composition of the training allocations changes. The design inherits from Queisler (2026c) the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits with their natural validation and test partitions, the nested round-robin patch order, and the readout: multinomial logistic regression with its regularization strength selected on the validation partition for every fit. It also inherits the patient distance. For split r and class c , $\bar{\mathbf{x}}_{irc}$ is the mean feature vector of patient i ’s patches of the class, $\boldsymbol{\mu}_r$ the average of these means over all classes, and

$$\mathbf{e}_{irc} = \frac{\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r}{\|\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r\|}, \quad d(i, j) = 1 - \mathbf{e}_{irc}^\top \mathbf{e}_{jrc}. \quad (1)$$

Table 1 lists the allocations. The deep and the broad random allocation are those of Queisler (2026c); only the low-coverage allocation differs.

2.1 Allocation rules

Three rules define the training sets and check the manipulation before any classifier is trained. First, every class is admitted. The pool $\mathcal{E}_{rc}$ contains the training patients with at least 32 patches of class c ; pools hold between 30 and 514 patients, so every class can supply twenty.

Second, the allocations are nested. For each split, class, and draw $d \in \{1, \dots, 5\}$, the five anchors are the patients of the stored five-patient draw of Queisler (2026a), whose predictions

Allocation	G	m	Patients
Deep	5	32	5 random anchors
Broad, concentrated	20	8	anchors, 15 neighbours of focal anchor
Broad, random	20	8	anchors, 15 random patients

Table 1: The three allocations share a budget of 160 patches per class and form two steps. G is the number of patients and m the number of patches per patient. From the deep to the broad concentrated allocation, patient count quadruples while all added patients resemble one anchor. From the broad concentrated to the broad random allocation, coverage increases while patients, patches per patient, budget, and effective support stay fixed.

are reused for the deep allocation. One anchor is drawn uniformly at random as the *focal anchor*. The concentrated allocation adds the fifteen patients of $\mathcal{E}_{rc}$ that are not anchors and lie closest to the focal anchor under Equation (1), with ties broken by patient identifier. The random allocation adds the same fifteen randomly drawn patients as in Queisler (2026c), so it is distributed like the twenty-patient draw of the breadth–depth grid. With $\mathcal{P}^D$, $\mathcal{P}^C$, and $\mathcal{P}^R$ the patient sets of the three allocations, $\mathcal{P}^D \subset \mathcal{P}^C$, $\mathcal{P}^D \subset \mathcal{P}^R$, and $|\mathcal{P}^C| = |\mathcal{P}^R| = 20$. An anchor’s eight patches in either broad allocation are the first eight of its 32 deep patches. The design requires 30 new fits: the broad random fits of Queisler (2026c) were never run, so both broad allocations are fitted for three splits and five draws.

Third, a manipulation check compares how well the allocations cover held-out patients. The *coverage distance* of a patient set is the mean distance from the class’s validation patients $\mathcal{V}_{rc}$ to their nearest training patient,

$$r_{rc}(\mathcal{P}) = \frac{1}{|\mathcal{V}_{rc}|} \sum_{i \in \mathcal{V}_{rc}} \min_{j \in \mathcal{P}} d(i, j). \quad (2)$$

With r_D , r_C , and r_R averaged over classes and draws within a split, the *coverage leakage*

$$\kappa = \frac{r_D - r_C}{r_D - r_R} \quad (3)$$

is the share of the random allocation’s coverage improvement that the concentrated allocation already attains. Fitting proceeds only if $\kappa \leq 0.5$ in all three splits; the class-level κ_c is recorded alongside. The rule targets this quantity directly: a validation patient whose nearest anchor is not the focal anchor can gain from the concentrated additions only if one of them lies closer to it than that anchor, which the concentration around another anchor makes unlikely. With one focal anchor among five, roughly a fifth of the validation patients can benefit.

3 Evaluation

Both steps of Table 1 are read with the same accuracy measure and the same uncertainty procedure.

3.1 Coverage gain and within-concentration residual

Accuracy is patient-macro balanced accuracy over all 30 classes: patch recall is averaged within each test patient and true class, then across patients within each class, and finally across classes. Averaging three splits and five draws gives A_D , A_C , and A_R in percent. The *coverage gain*

$$\Delta_C = A_R - A_C \quad (4)$$

measures what dissimilar patients contribute at fixed patients, patches per patient, budget, and effective support. Under patient coverage it is positive; under count beyond coverage it is close to zero.

The first step also raises effective support. With $\bar{\rho}_c$ the full-feature intraclass correlation of class c (Queisler, 2026b), effective support is $N_{\text{eff}}(G, m) = \frac{1}{30} \sum_c Gm / (1 + (m - 1)\bar{\rho}_c)$, which gives 10.06 per class for the deep and 36.58 for both broad allocations. The surface $A = \alpha + \beta \log N_{\text{eff}} + \gamma \log G$ is fitted to the stored predictions of the nine-cell breadth–depth grid, and its reference residual $b_{\text{ref}} = \hat{\gamma} \log 2$ must reproduce 2.62 points per doubling. Dividing the unexplained part of the first step by its two doublings of patients gives the *within-concentration residual*

$$b_C = \frac{A_C - A_D - \hat{\beta} \log(N_{\text{eff}}(20, 8)/N_{\text{eff}}(5, 32))}{2} \quad (5)$$

in points per doubling of patients. If coverage carries the residual, b_C falls well below b_{ref} and may turn negative, because effective support counts the concentrated patients as independent of each other. If coverage does not matter, b_C stays close to b_{ref} .

Two secondary analyses describe where a coverage gain arises. First, each test patient is assigned per draw to the *focal cell* if its nearest anchor is the focal anchor and to the *other cells* otherwise, and Δ_C is computed within both strata with the patient weights of the primary endpoint. Patient coverage predicts a gain concentrated in the other cells, which only the random allocation extends, and no gain or a loss in the focal cell; count beyond coverage predicts similar gains in both. Second, all quantities are reported separately for the thirteen site classes of Queisler (2026d) and the seventeen remaining classes, with the surface refitted to each subgroup, together with the share of concentrated and random additions that come from the focal anchor’s tissue source site. These analyses carry no decision threshold.

3.2 Uncertainty

Uncertainty follows Queisler (2026a). Each of 2,000 paired Bayesian bootstrap replicates assigns Dirichlet(1, ..., 1) weights to the distinct test patients (Rubin, 1981), shared across allocations, splits, draws, and classes. The same weights enter the refitted surface, so $\hat{\beta}$, b_{ref} , and b_C remain paired. The 2.5th and 97.5th percentiles give exploratory 95% intervals, including one for $b_C - b_{\text{ref}}$. Correlations and embeddings are held fixed. The standard deviation of each allocation’s accuracy and of both steps across the fifteen split–draw combinations is reported as a descriptive measure of sensitivity to the drawn anchors.

4 Interpretation criteria

Conclusions rest on the test-patient intervals for Δ_C and b_C and on the practical threshold of one percentage point used in the preceding experiments (Table 2).

Patient coverage. If dissimilar patients add a practically relevant gain and concentrated patients add nothing beyond effective support, the missing information on TCGA-UT is coverage of between-patient variation. Coverage distance would then complement effective support as a shortage signal, and a mitigation would select or weight patients for annotation by their distance to the patients already covered.

Count beyond coverage. If dissimilar patients add nothing while the within-concentration residual stays above one point, the benefit of additional patients does not depend on how different their mean appearance is. The missing information would lie in properties a patient mean does not express, such as the distribution of a patient’s patches, and a signal would need a distance between patch distributions.

Account	Claim	Coverage gain Δ_C	Within-concentration residual b_C
Patient coverage	Additional patients help because they differ from the patients already in training	Interval above 1	Interval below 1
Count beyond coverage	Additional patients help even when they resemble the patients already in training	Interval within $[-1, 1]$	Interval above 1
Both	Coverage and patient count each contribute	Interval above 1	Interval above 1

Table 2: Each account predicts a distinct combination of the two contrasts. Intervals are test-patient 95% intervals in percentage points; b_C is expressed per doubling of patients. The coverage account admits a negative b_C . All other combinations, including any interval that crosses a threshold, are inconclusive.

Both and inconclusive outcomes. If both contrasts exceed the threshold, coverage explains part of the residual, and $b_C - b_{\text{ref}}$ quantifies which part; the leakage κ bounds how much coverage the concentrated allocation still adds. All other combinations leave the accounts open.

5 Results

The manipulation check ran for all 30 cancer types, three splits, and five draws per class, and it stopped the experiment before any classifier was trained. Coverage leakage is $\kappa = 0.69, 0.67$, and 0.73 in the three splits, against a permitted maximum of 0.5 . Concentrating all additions around one focal anchor lowers leakage from the 0.84 to 0.88 of the neighbour allocation (Queisler, 2026c), but the concentrated allocation still attains about two thirds of the coverage improvement of the random allocation. The coverage gain Δ_C and the within-concentration residual b_C are therefore not estimated, and the accounts of Table 2 remain untested. The check does measure where the coverage improvement arises, and a diagnostic on the same embeddings, reported in Sections 5.3 and 5.4, tests the design’s central premise and several alternative allocation rules before any further fitting.

5.1 Manipulation check

The concentrated allocation shortens the coverage distance of held-out patients by clearly less than the random allocation, but by far more than half as much (Table 3). Averaged over classes and draws, validation patients lie at a mean distance of 0.386 from their nearest patient of the deep allocation. Adding fifteen neighbours of the focal anchor shortens this distance to 0.322 , and adding fifteen random patients shortens it to 0.294 . The room between the two broad allocations thus grows from 0.013 in the neighbour design to 0.028 , yet the concentrated allocation still delivers a reduction of 0.062 to 0.068 . The three splits agree closely.

Split	r_D	r_C	r_R	$r_D - r_R$	κ
1	0.387	0.324	0.296	0.091	0.690
2	0.385	0.324	0.293	0.092	0.668
3	0.385	0.317	0.292	0.093	0.734

Table 3: The concentrated allocation attains about two thirds of the coverage improvement of the random allocation, so the manipulation check of Section 2.1 fails in every split. Coverage distances are mean cosine distances from validation patients to their nearest training patient, averaged over the 30 classes and five draws; κ is the coverage leakage of Equation (3), which the design requires to be at most 0.5.

5.2 Leakage by cancer type

Concentration lowers leakage in almost every cancer type, but only one reaches the intended contrast. Averaged over splits, class-level leakage has a median of 0.69, down from 0.86 for the neighbour allocation, and ranges from 0.38 for testicular germ cell tumours to 1.04 for uveal melanoma (Table 4). Across the 90 combinations of classes and splits, six fall to 0.5 or below: testicular germ cell tumours in all three splits, and rectum adenocarcinoma, lung adenocarcinoma, and thyroid carcinoma in one split each. Six combinations exceed one: uveal melanoma and uterine carcinosarcoma in two splits each, and ovarian serous cystadenocarcinoma and rectum adenocarcinoma in one split each. Leakage stays high in small and large classes alike, as it did for the neighbour allocation.

Concentration restricts institutions only mildly. On average, 27 percent of the concentrated additions share the focal anchor’s tissue source site, against 17 percent of the random additions. Two classes with few contributing sites, adrenocortical carcinoma and pheochromocytoma and paraganglioma, reach about 70 percent under both rules, so the difference between the rules there is nil. The overlap with site coverage that Section 7 anticipates is thus small compared with the failure of the coverage manipulation.

5.3 Why concentration does not withhold coverage

The design rests on one premise: a validation patient whose nearest anchor is not the focal anchor should gain nothing from the concentrated additions. A diagnostic on the same embeddings, draws, and allocations tests this premise directly. It reproduces the coverage distances of Table 3 exactly and splits each allocation’s improvement $r_D - r$ into the contributions of validation patients in the focal cell and in the other cells, each expressed as a share of the random allocation’s total improvement $r_D - r_R$ (Table 5).

The focal cell behaves as intended but is small. It holds 17 to 23 percent of the validation patients, close to the one fifth expected from five anchors, and concentrated additions improve coverage there only slightly more than random additions do, contributing 21 to 29 percent of the random allocation’s improvement against 18 to 25 percent. The premise fails in the other cells. For 55 to 57 percent of the validation patients outside the focal cell, one of the fifteen neighbours of the focal anchor lies closer than their own nearest anchor, and these patients contribute 43 to 48 percent of the random allocation’s improvement, about twice the contribution of the focal cell. Leakage thus arises mainly where the design expected none.

The reason is the scale of the anchor cells. Five randomly drawn anchors leave validation patients at a mean distance of 0.39 from their nearest anchor, so a cell is not a tight neighbourhood but a large share of the class. The fifteen patients nearest to one anchor spread over a region that lies closer to many validation patients than their own anchor does. The same diagnostic shows how weakly the coverage distance separates cohorts by composition: twenty patients consisting of the focal anchor and its nineteen nearest neighbours cover validation patients at 0.346

Cancer type	Coverage distance			κ_c	Focal-site share, %	
	r_D	r_C	r_R		Concentrated	Random
Adrenocortical carcinoma	0.309	0.258	0.246	0.81	72	71
Bladder urothelial carcinoma	0.573	0.479	0.430	0.66	12	6
Brain lower grade glioma	0.261	0.219	0.196	0.65	24	8
Breast invasive carcinoma	0.541	0.468	0.430	0.66	18	11
Cervical squamous and endocervical carcinoma	0.521	0.442	0.383	0.57	11	10
Colon adenocarcinoma	0.310	0.270	0.245	0.62	24	5
Esophageal carcinoma	0.560	0.466	0.410	0.62	22	7
Glioblastoma multiforme	0.324	0.270	0.245	0.69	45	13
Head and neck squamous cell carcinoma	0.359	0.301	0.274	0.69	21	12
Kidney chromophobe	0.203	0.159	0.153	0.86	18	15
Kidney renal clear cell carcinoma	0.256	0.218	0.191	0.59	36	18
Kidney renal papillary cell carcinoma	0.355	0.297	0.276	0.73	9	3
Liver hepatocellular carcinoma	0.368	0.309	0.285	0.71	32	19
Lung adenocarcinoma	0.500	0.429	0.385	0.62	21	7
Lung squamous cell carcinoma	0.499	0.424	0.394	0.71	16	5
Mesothelioma	0.512	0.421	0.398	0.79	8	5
Ovarian serous cystadenocarcinoma	0.441	0.346	0.341	0.95	56	44
Pancreatic adenocarcinoma	0.438	0.356	0.344	0.87	13	6
Pheochromocytoma and paraganglioma	0.249	0.183	0.161	0.76	68	69
Prostate adenocarcinoma	0.189	0.167	0.153	0.61	16	9
Rectum adenocarcinoma	0.272	0.221	0.199	0.70	20	13
Sarcoma	0.379	0.325	0.287	0.59	40	26
Skin cutaneous melanoma	0.493	0.420	0.396	0.76	20	11
Stomach adenocarcinoma	0.564	0.461	0.416	0.70	27	16
Testicular germ cell tumours	0.274	0.245	0.198	0.38	48	39
Thymoma	0.318	0.260	0.223	0.61	13	9
Thyroid carcinoma	0.341	0.282	0.237	0.56	21	17
Uterine carcinosarcoma	0.483	0.397	0.395	0.98	17	13
Uterine corpus endometrial carcinoma	0.439	0.375	0.339	0.64	16	3
Uveal melanoma	0.239	0.179	0.182	1.04	35	25
Mean over classes	0.386	0.322	0.294	0.70	27	17

Table 4: Class-level leakage stays above the permitted 0.5 for every cancer type except testicular germ cell tumours. Coverage distances and leakage are averaged over the three splits and five draws; κ_c is computed from the class’s own coverage distances. The focal-site shares give the percentage of added patients that come from the tissue source site of the focal anchor, for the concentrated and the random addition rule.

Split	Focal cell, %	Concentrated, %		Random, %		Other cells closer, %
		Focal	Other	Focal	Other	
1	17	21	48	18	82	57
2	21	24	43	20	80	56
3	23	29	45	25	75	55

Table 5: Most of the concentrated allocation’s coverage improvement comes from validation patients outside the focal cell, which the design assumed it would not reach. The focal cell holds the validation patients whose nearest anchor is the focal anchor. The middle columns give each cell’s contribution to an allocation’s coverage improvement as a percentage of the random allocation’s total improvement; the concentrated columns sum to 100%. The last column gives the percentage of other-cell validation patients whose nearest training patient is closer in the concentrated than in the deep allocation.

to 0.356, better than the five random anchors at 0.385. Under the patient-mean embedding, adding almost any patients shortens the nearest-patient distance for many held-out patients.

5.4 Alternative allocation rules

Because each manipulation check costs no classifier fits, the diagnostic also evaluated alternative allocation rules against the same criterion before any redesign is fixed. Two coverage-maximizing rules select from the training pool without using validation patients: *farthest-point selection* repeatedly adds the pool patient farthest from the current set, and *facility-location selection* repeatedly adds the pool patient that most lowers the mean distance from all pool patients to their nearest selected patient. For an upward contrast, the random allocation is the low arm, and the criterion requires it to attain at most half of the high arm’s improvement. A *fully concentrated* variant replaces the random anchors by the focal anchor and its four nearest neighbours, so that the deep allocation is itself concentrated and must be fitted anew. Table 6 lists the leakage of every variant.

Three observations constrain any redesign. First, farthest-point selection covers held-out patients worse than random selection, with leakage above one, because it adds outlying patients whom few validation patients resemble. Second, facility-location selection covers better than random selection, at 0.269 to 0.273 against 0.292 to 0.296, but random selection still attains 80 percent of its improvement, so an upward contrast is as weak as the downward ones. Third, combining the concentrated and the facility-location arm roughly halves the leakage of the neighbour design but still stops at 0.53 to 0.59. All seven variants build a twenty-patient allocation on top of a five-patient one, and within this nested structure the step from five to twenty patients dominates every coverage change that the selection rule can add.

6 Discussion and conclusion

Whether the residual breadth benefit of TCGA-UT is a patient-coverage benefit remains open after a second stopped manipulation check. Concentrating all additions around one anchor lowers leakage from 0.84–0.88 to 0.67–0.73, but no classifier was trained, so neither the coverage gain nor the within-concentration residual constrains the accounts of Table 2.

The two checks together yield a result about the design space. The diagnostic refutes the assumption on which the concentrated design rests: anchor cells do not localize coverage, and validation patients outside the focal cell supply about twice as much of the concentrated allocation’s improvement as those inside it. Across seven nested variants, including upward contrasts

Low-coverage arm	High-coverage arm	Leakage per split			Cells ≤ 0.5
Anchors + neighbours of all anchors	Anchors + random	0.84	0.85	0.88	0 of 90
Anchors + neighbours of focal anchor	Anchors + random	0.69	0.67	0.73	6 of 90
Anchors + random	Anchors + farthest-point	1.39	1.35	1.37	0 of 90
Anchors + random	Anchors + facility-location	0.80	0.80	0.80	1 of 90
Anchors + neighbours of focal anchor	Anchors + facility-location	0.55	0.53	0.59	36 of 90
Focal anchor + 19 neighbours	Focal anchor + 4 neighbours + random	0.64	0.61	0.66	13 of 90
Focal anchor + 19 neighbours	Focal anchor + 4 neighbours + facility-location	0.56	0.53	0.57	32 of 90

Table 6: No nested allocation rule reaches the permitted leakage of 0.5 in any split. Leakage is the share of the high arm’s coverage improvement over the five-patient allocation that the low arm already attains, computed as in Equation (3) from coverage distances averaged over classes and draws. The first two rows are the neighbour design of Queisler (2026c) and the design of this report. Cells count class–split combinations at or below 0.5.

with coverage-maximizing rules, the best leakage is 0.53. Under the patient-mean embedding, any nested design that must add little coverage while quadrupling patients runs against the fact that quadrupling patients shortens the nearest-patient distance for most held-out patients, whichever patients are added. A clean within-concentration residual, which requires a step that raises patient count at nearly constant coverage, is therefore out of reach for this coverage measure.

The measure is not insensitive to composition, however. Its dynamic range between cohorts of equal size is large once no five-patient allocation has to be nested inside both arms: in the diagnostic, twenty patients forming one tight neighbourhood cover validation patients at 0.346 to 0.356, and twenty patients selected by facility location starting from the same focal anchor cover them at 0.268 to 0.271. This difference is almost as large as the coverage change from five to twenty random patients. The question whether dissimilar patients help can thus still be asked directly, at a fixed patient count, even though the question whether similar patients help beyond effective support cannot be asked with nested allocations.

6.1 Follow-up: coverage, random sampling, and learning

A follow-up experiment would compare three allocations of twenty patients contributing eight patches each: the focal anchor and its nineteen nearest neighbours (*clustered*), nineteen randomly drawn additions (*random*), or nineteen additions selected by facility location on the training pool (*dispersed*). All three share the focal anchor, patient count, patch budget, and patch-sampling rule. The random allocation distinguishes damage from deliberate concentration from improvement over ordinary sampling. Three splits and five draws require 45 fits, with the same logistic-regression readout and validation tuning as this experiment.

The clustered–dispersed coverage contrast is already substantial: it reaches 0.90, 0.95, and 0.84 times the coverage change from five to twenty random patients. The follow-up would retain a minimum ratio of 0.5 in every split as its manipulation check, while reporting class-level contrasts, especially for the seventeen smaller cancer types. These diagnostics establish feasibility, not an expected accuracy gain. Before fitting, the coverage measure and analysis would be fixed; an exploratory analysis of stored random allocations would test whether coverage predicts ac-

curacy among draws with identical patient counts and patch depths, and adds predictive value beyond effective support and patient count.

The primary contrasts would be dispersed minus random accuracy and random minus clustered accuracy, using patient-macro balanced accuracy, paired test-patient intervals, and the one-percentage-point threshold. A gain over random sampling would support coverage-guided selection; a gain only over clustering would establish concentration damage without demonstrating improved ordinary sampling. An interval within $[-1, 1]$ would indicate practical equivalence for the corresponding contrast at this allocation size, whereas an interval crossing a threshold would remain inconclusive. Training-draw sensitivity and the association between each test patient’s coverage improvement and recall change would be reported alongside, separately for the thirteen site classes and the seventeen remaining classes.

Equal calculated effective support does not ensure equal dependence between selected patients, and changes in site composition or within-class frequencies can also affect accuracy. The contrasts therefore test selection strategies without uniquely identifying coverage as the cause of the original breadth benefit. If coverage predicts ordinary-cohort performance and selection improves it, a subsequent comparison of coverage-selected and random five-patient cohorts at the same patch budget would test mitigation under shortage. A null result would weaken patient-mean coverage at the tested size, rather than exclude coverage generally; diagnostically relevant variation in patients’ patch distributions would remain a candidate.

7 Limitations

The concentrated allocation does more than withhold coverage. Sixteen of its twenty patients resemble one anchor, so it also over-represents one appearance of the cancer type, and a classifier may shift its class boundary towards that appearance. A positive coverage gain can therefore reflect this imbalance within the class as well as missing coverage. The focal-cell strata separate the two only partly: both readings predict a gain in the other cells.

Neighbours of one anchor may share its tissue source site, because site signatures are visible in deep features of TCGA slides (Howard et al., 2021). The reported site shares quantify this overlap, and the site gain of 1.06 points (Queisler, 2026d) gives a reference scale for how much of a coverage gain sites alone could explain.

Coverage is defined on patient means in one representation, so a null coverage gain excludes coverage only as measured here. The within-concentration residual inherits the effective-support surface, whose slope is estimated on random allocations and transferred to a concentrated one. All results are conditional on frozen Virchow2 features, multinomial logistic regression, the three overlapping splits, and the tested allocation sizes.

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